Category Theory for Mathematical Physics — Exercise Sheet 12

Morita equivalence and the Drinfeld centre

Variant: Hints and answers

Conventions as in Lecture 12: a half-braiding is $\gamma_X \colon X \otimes Z \to Z \otimes X$, and the braiding of $\mathcal{Z}(C)$ is $c_{(Z,\gamma),(Z',\gamma')} = \gamma'_Z$ (Theorem 12.7). The toric code's anyons are $1 = (0,1)$, $e = (0,\chi_-)$, $m = (1,1)$, $\varepsilon = (1,\chi_-)$ in $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ (Theorem 12.8), with $\mathbb{Z}_2 = \{0,1\}$ written additively and $\chi_-(h) = (-1)^h$; here e is an anyon, and the identity of S_3 is written 1. $\zeta = e^{2\pi i/3}$, $\varphi = (1+\sqrt{5})/2$, and ω is a 3-cocycle (Exercise 6). S, T and the central charge c are those of Lecture 10.

Exercise 1 (The S-matrix of the toric code from half-braidings). (i) Draw the condition of Definition 12.6 for a half-braiding on $Z = \delta_g$ in $\mathrm{Vec}_{\mathbb{Z}_2}$, with $X = \delta_h$ and $Y = \delta_k$, as in Equation (12.1). Writing $\gamma_{\delta_h} = \gamma(h)\,\mathrm{id}$, show that γ is a character of $\mathbb{Z}_2$, and recover the four simple objects (g,χ) of Theorem 12.8.

(ii) Draw $c_{m,e}$ as in Equation (12.2), and show that the braiding is $c_{X,Y} = \chi_Y(g_X)$. Write the table of $c_{X,Y}$ and of the double braidings $c_{Y,X}c_{X,Y}$ for $X, Y \in \{1, e, m, \varepsilon\}$.

(iii) Using that every d is 1 and every anyon is its own dual, show that $\tilde{S}_{XY}$ is the double braiding, and compare $S = \tilde{S}/2$ with Example 10.14.

(iv) Compute T and the central charge.

Hint. (i) The associators of $\mathrm{Vec}_{\mathbb{Z}_2}$ are trivial. (iii) The Hopf link of two invertible objects is their double braiding times $d_X d_Y$.

Answer. (i) $\gamma(h+k) = \gamma(h)\gamma(k)$ and $\gamma(0) = 1$: $\gamma = \chi_\pm$, two half-braidings on each of δ_0, δ_1. (ii) $c_{m,e} = \gamma_e(\delta_1) = \chi_-(1) = -1$. The table has rows $(1,1,1,1)$, $(1,1,1,1)$, $(1,-1,1,-1)$, $(1,-1,1,-1)$; the double braiding is -1 exactly for the pairs among e, m, ε that are different. (iii) $S = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{smallmatrix}\right)$, as in Lecture 10. (iv) $T = \mathrm{diag}(1,1,1,-1)$; Gauss sum 2, $c = 0$.

Exercise 2 (The double $D(G)$, and $G = \mathbb{Z}_3$). (i) Show that $D(G)$ (Theorem 12.9) is associative, with unit $\sum_g \delta_g \otimes 1$, and has dimension $|G|^2$.

(ii) Let G be abelian. Show that $D(G)$ is commutative, and that for $a \in G$ and a character χ of G, $\rho_{a,\chi}(\delta_g \otimes h) = \delta_{a,g}\,\chi(h)$ is an algebra map $D(G) \to \mathbb{C}$.

(iii) For $G = \mathbb{Z}_3$, list the nine irreducible representations and their spins $\theta = \chi(a)$, and compare with Theorem 12.8.

(iv) Label the simple objects of $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_3})$ by (a,b), with flux a and character $\chi_b(h) = \zeta^{bh}$. Using Definition 10.2 and the braiding of Theorem 12.8, show that $S_{(a,b),(a',b')} = \zeta^{-(ab'+a'b)}/3$. For invertible S and T, $(ST)^3 = S^2$ is equivalent to $STS = T^{-1}ST^{-1}$: check the entry at $X = (0,1)$, $Y = (1,0)$, and show that it fails there for the complex conjugate $\bar{S}$ with the same T.

Hint. (i) Multiply three basis elements both ways; the Kronecker deltas must agree. (ii) For abelian G, $hg'h^{-1} = g'$. (iii) An irreducible representation of a commutative algebra over $\mathbb{C}$ is one-dimensional (Schur). (iv) $\bar{Y} = (-a', -b')$; for the entry, $ab + a + b = (a+1)(b+1) - 1$.

Answer. (i) Both products of $\delta_g \otimes h$, $\delta_{g'} \otimes h'$, $\delta_{g''} \otimes h''$ are $\delta_{g,hg'h^{-1}}\delta_{g',h'g''h'^{-1}}\,\delta_g \otimes hh'h''$. (ii) The product is $\delta_{g,g'}\,\delta_g \otimes hh'$, symmetric; $\rho_{a,\chi}$ multiplies the deltas and the characters. (iii)

$\rho_{a,b}$ with $\chi_b(h) = \zeta^{bh}$; spins ζ^{ab}: five equal to 1, two to ζ, two to ζ^2; the labels (g,χ) of Lecture 12. (iv) Both sides are $\zeta^2/3$ for S; for $\bar{S}$ the left side is $\zeta^2/3$ and the right side $\zeta/3$.

Exercise 3 (The eight anyons of $\mathcal{Z}(\mathrm{Vec}_{S_3})$). (i) List the conjugacy classes of S_3 and their centralisers, and the simple objects $([g],\pi)$ of Theorem 12.9.

(ii) The dimension of $([g],\pi)$ is $|[g]|\dim\pi$. Check that the squares add up to $|S_3|^2$.

(iii) Compute the spins $\theta = \chi_\pi(g)/\dim\pi$.

Hint. (i) The centraliser of a transposition has order $6/3$. (ii) A simple object of the centre is a sum of $\delta_{g'}$ over the class, each with the space of π.

Answer. (i) $\{1\}$ with S_3 (irreps $\mathbf{1}$, sgn, std); transpositions with $\mathbb{Z}_2$ (two); 3-cycles with $\mathbb{Z}_3$ (three): $3+2+3=8$. (ii) $1+1+4+9+9+4+4+4=36$. (iii) $1,1,1$; $1,-1$; $1,\zeta,\zeta^2$.

Exercise 4 (Rep($\mathbb{Z}_2$), and the symmetry $e\leftrightarrow m$). (i) Following the proof of Theorem 12.2(ii), find the simple module endofunctors (V,s) of $\mathrm{Vect}_{\mathrm{fd}}$ over $\mathrm{Vec}_{\mathbb{Z}_2}$ and their composition, and show that $(\mathrm{Vec}_{\mathbb{Z}_2})^*_{\mathrm{Vect}_{\mathrm{fd}}} \simeq \mathrm{Rep}(\mathbb{Z}_2) \simeq \mathrm{Vec}_{\mathbb{Z}_2}$.

(ii) Show that the exchange $e\leftrightarrow m$ (fixing 1 and ε) preserves the fusion rules, the spins and the double braidings of the toric code. Does it preserve each $c_{X,Y}$?

(iii) Write each anyon as $X=(g_X,b_X)\in\mathbb{Z}_2\times\mathbb{Z}_2$ with $\chi = \chi_-^{b_X}$; the toric code has trivial associators. Let F be the exchange $(g,b)\mapsto(b,g)$ on simple objects, extended linearly, with $J_{X,Y} = (-1)^{g_X b_Y}$ and $J_0 = 1$. Show that (F,J,J_0) is a monoidal functor (Definition 3.10) and that it is *braided*, $F(c_{X,Y})\circ J_{X,Y} = J_{Y,X}\circ c_{FX,FY}$: the exchange is a braided autoequivalence.

Hint. (i) $s_{g+h} = s_h s_g$ on a line. (ii) Use the tables of Exercise 1. (iii) With trivial associators the condition on J is a cocycle condition on scalars.

Answer. (i) $V=\mathbb{C}$ with $s_1=\pm1$: the trivial and the sign representation, multiplying like $\mathbb{Z}_2$, with trivial associator. (ii) Yes for fusion ($e\otimes m=\varepsilon$ is symmetric), spins $(1,1)$ and double braidings; no: $c_{e,m}=1$, $c_{m,e}=-1$. (iii) J is bilinear, hence a cocycle, and $J_{X,Y}=c_{X,Y}$, so both sides of the braided condition are 1.

Exercise 5 (Centres of modular categories). You may use that on $\mathcal{C}\boxtimes\mathcal{D}$ the braiding, the dimensions and the spins are taken factor by factor, $d_{a\boxtimes b}=d_a d_b$ and $\theta_{a\boxtimes b}=\theta_a\theta_b$, and that $\mathcal{C}^{\mathrm{rev}}$ has the spins $\bar{\theta}_a$.

(i) By Theorem 12.10(iii), $\mathcal{Z}(\mathrm{Fib}) \simeq \mathrm{Fib}\boxtimes\mathrm{Fib}^{\mathrm{rev}}$. List its simple objects with their dimensions and spins, and check $\dim\mathcal{Z}(\mathrm{Fib}) = (1+\varphi^2)^2$.

(ii) Show that for any unitary modular $\mathcal{C}$ the Gauss sum of $\mathcal{C}\boxtimes\mathcal{C}^{\mathrm{rev}}$ is $\dim\mathcal{C}$, so $c=0$.

(iii) How many simple objects does $\mathcal{Z}(\mathrm{SU}(2)_k)$ have, and what is its dimension?

Hint. (ii) Write the double sum as a product of two sums.

Answer. (i) $(1,1),(\tau,1),(1,\tau),(\tau,\tau)$ with $d=1,\varphi,\varphi,\varphi^2$ and $\theta=1,e^{4\pi i/5},e^{-4\pi i/5},1$; dim $=1+2\varphi^2+\varphi^4=(1+\varphi^2)^2$. (ii) $\sum_{a,b} d_a^2 d_b^2\theta_a\bar{\theta}_b = |p_+|^2 = \dim\mathcal{C}$, real and positive. (iii) $(k+1)^2$ simple objects, dimension $\big((k+2)/(2\sin^2(\pi/(k+2)))\big)^2$.

Exercise 6 ⋆. *The double semion.* Let $\mathcal{C}=\mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$, $\mathbb{Z}_2=\{0,1\}$, with $\omega(1,1,1)=-1$, the other values 1.

(i) For a half-braiding on δ_g, write $\gamma_{\delta_h}=\gamma(h)\,\mathrm{id}$ with $\gamma(0)=1$. Show that Definition 12.6 becomes

$$\gamma(h+k) = \frac{\omega(g,h,k)\,\omega(h,k,g)}{\omega(h,g,k)}\,\gamma(h)\gamma(k).$$

(ii) Solve it: find the four simple objects of $\mathcal{Z}(\mathcal{C})$ and their spins.

(iii) Conclude with Theorem 12.11 that $\mathrm{Vec}_{\mathbb{Z}_2}$ and $\mathcal{C}$ are not Morita equivalent.

Hint. (i) On simple objects every arrow in the half-braiding condition is a scalar. (ii) Use that ω is normalised, then look at $h = k = 1$. (iii) The spin of an invertible object is its self-braiding.

Answer. (i) The associators contribute $\omega(g,h,k)$, $\omega(h,g,k)^{-1}$ and $\omega(h,k,g)$. (ii) $g = 0$: $\gamma(1) = \pm 1$; $g = 1$: $\gamma(1)^2 = -1$, $\gamma(1) = \pm i$; spins $1, 1, i, -i$. (iii) The toric code's self-braidings are $1, 1, 1, -1$: no braided equivalence matches them.